

Gradient descent manual calculation

$$y = mx + b, \quad \boxed{m = -1}, \quad \boxed{b = 1}; \quad \boxed{\alpha = 0.1}$$

Given points: (1, 3) and (3, 6)

$$\rightarrow \frac{dJ}{dm} = -\frac{2}{n} \sum (y_i - \hat{y}_i) x_i$$

$$\rightarrow \frac{dJ}{db} = -\frac{2}{n} \sum (y_i - \hat{y}_i), \quad n = \text{number of data points.}$$

$$\rightarrow m_{\text{new}} = m_{\text{old}} - \alpha \frac{dJ}{dm}$$

$$\rightarrow b_{\text{new}} = b_{\text{old}} - \alpha \frac{dJ}{db}$$

1. Compute the predicted values $\hat{y}$ for each data point

$$\text{for } (x_1, y_1) = (1, 3), \quad \hat{y}_1 = m \cdot 1 + b \\ = (-1)(1) + 1 = -1 + 1 = 0$$

$$\boxed{\hat{y}_1 = 0}$$

$$\text{for } (x_2, y_2) = (3, 6), \quad \hat{y}_2 = m \cdot 3 + b \\ = (-1)(3) + 1 = -3 + 1 = -2$$

$$\boxed{\hat{y}_2 = -2}$$

2. Calculate partial derivatives.

$$\frac{dJ}{dm} = -\frac{2}{n} [(y_1 - \hat{y}_1) x_1 + (y_2 - \hat{y}_2) x_2], \quad (y_1 - \hat{y}_1) = (3 - 0) = 3$$

$$\frac{dJ}{db} = -\frac{2}{n} [(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2)], \quad (y_2 - \hat{y}_2) = 6 - (-2) = 8$$

$$\frac{dJ}{dm} = -\frac{2}{2} [(3)(1) + (8)(3)] = -1[3 + 24] = -27$$

$$\frac{dJ}{db} = -\frac{2}{2} [(3) + (8)] = -1[11] = -11$$

3. Update $m \times b$

$$m_{\text{new}} = -1 - (0.1)(-27) \\ = -1 + 2.7 = 1.7, \quad \boxed{m_{\text{new}} = 1.7}$$

$$b_{\text{new}} = 1 - (0.1)(-11)$$

$$b_{\text{new}} = 1 - (0.1)(-11) = 1 + 1.1 = 2.1, \quad \boxed{b_{\text{new}} = 2.1}$$

Convert to $\bar{y}$: $m = 1.7$, $b = 2.1$, $\sigma^2 = 0.1$

$$\text{Data point} = (x_1, y_1) = (1, 3) \\ (x_2, y_2) = (3, 6)$$

$$\boxed{y = mx + b}$$

$$y_1 = 1.7(1) + 2.1 = 3.8 \quad , \quad \boxed{y_1 = 3.8}$$

$$y_2 = 1.7(3) + 2.1 = 7.2 \quad , \quad \boxed{y_2 = 7.2}$$

$$\frac{dy}{dx} = -\frac{n}{2} \left[(y_1 - \hat{y}_1) x_1 + (y_2 - \hat{y}_2) x_2 \right]$$

$$= -\frac{2}{2} \left[(3 - 3.8)(1) + (6 - 7.2)(3) \right]$$

$$= (-1) (-0.8 - 3.6)$$

$$= -1 (-4.4)$$

$$= \underline{\underline{4.4}}$$

$$\frac{dy}{dt} = -\frac{1}{2} [(y_1 - y_1') + (y_2 - y_2')]$$

$$= -\frac{1}{2} [(3.38) + (6.72)]$$

$$= -1 (-0.8 - 1.2)$$

$$= -1(2)$$

$$= 2 //$$

$$\begin{aligned} m_{\text{new}} &= m_{\text{old}} - \Delta \frac{dy}{dm} = 1.7 - (0.1)(4.4) \\ &= 1.7 - 0.44 \\ &= 1.26 \end{aligned}$$

$$m_{\text{new}} = 1.26$$

$$\begin{aligned} b_{\text{new}} &= b_{\text{old}} - \Delta \frac{dy}{db} = 2.1 - 0.1(2) \\ &= 2.1 - 0.2 \\ &= 1.9 \end{aligned}$$

$$b_{\text{new}} = 1.9$$

Current values: $m = 126$, $b = 1.9$, $\alpha = 0.1$

Data points: $(x_1, y_1) = (1, 3)$

$(x_2, y_2) = (3, 6)$

$$y = mx + b$$

$$\hat{y}_1 = 126(1) + 1.9 = 3.16$$

$$\hat{y}_1 = 3.16$$

$$\hat{y}_2 = 126(3) + 1.9 = 5.68$$

$$\hat{y}_2 = 5.68$$

$$\begin{aligned} \frac{dy}{dm} &= -\frac{2}{2} [(y_1 - \hat{y}_1)x_1 + (y_2 - \hat{y}_2)x_2] \\ &= -\frac{2}{2} [(3 - 3.16)(1) + (6 - 5.68)(3)] \\ &= -1(-0.16 + 0.96) \\ &= -0.8 \end{aligned}$$

$$\begin{aligned}\frac{d\hat{y}}{db} &= \frac{-h}{2} \left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) \right] \\ &= -\frac{2}{2} \left[(3 - 3.16) + (6 - 5.68) \right] \\ &= -1(-0.16 + 0.32) \\ &= -0.16\end{aligned}$$

$$\begin{aligned}m_{\text{new}} &= m_{\text{old}} - \alpha \frac{dJ}{dm} \\ &= 1.26 - (0.1)(0.8) \\ &= 1.26 + 0.08 \\ &= \underline{1.34}\end{aligned}$$

$$m_{\text{new}} = 1.34$$

$$\begin{aligned}b_{\text{new}} &= b_{\text{old}} - \alpha \frac{dJ}{db} \\ &= 1.9 - (0.1)(0.16) \\ &= 1.9 + 0.016 \\ &= 1.916\end{aligned}$$

$$b_{\text{new}} = 1.916$$

Iteration 1: $m = 1.7, b = 2.1$

Iteration 2: $m = 1.26, b = 1.9$

Iteration 3: $m = 1.34, b = 1.916$

The changes in m & b are becoming smaller in magnitude, the values of m , and b are indeed moving towards reducing the error: iteration 1: $dJ/dm = -27, dJ/db = -11$

iteration 2: $dJ/dm = 4.4, dJ/db = 2$
iteration 3: $dJ/dm = -0.8, dJ/db = -0.1$

Iteration 4:

Current values: $m = 1.34$

$b = 1.916, \alpha = 0.1$

Data points: $(x_1, y_1) = (1, 3)$

$(x_2, y_2) = (3, 6)$

$y = mx + b$

$\hat{y}_1 = 1.34(1) + 1.916 = 3.256, \boxed{\hat{y}_1 = 3.256}$

$\hat{y}_2 = 1.34(3) + 1.916 = 5.936, \boxed{\hat{y}_2 = 5.936}$

$\frac{dJ}{dm} = -\frac{1}{2} [(y_1 - \hat{y}_1)x_1 + (y_2 - \hat{y}_2)x_2]$

$= -\frac{1}{2} [(3 - 3.256)(1) + (6 - 5.936)(3)]$

$= -1(-0.256 + 0.192)$

$= -1(-0.064)$

$= 0.064$

$\frac{dJ}{dm} = -\frac{1}{2} [(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2)]$

$= -\frac{1}{2} [(3 - 3.256) + (6 - 5.936)]$

$= -1(-0.256 + 0.064)$

$= -1(-0.192) = 0.192$

$m_{\text{new}} = m_{\text{old}} - \alpha \cdot \frac{dJ}{dm} = 1.34 - 0.1(0.064) = 1.34 - 0.0064 = 1.3336$

$b_{\text{new}} = b_{\text{old}} - \alpha \cdot \frac{dJ}{db} = 1.916 - 0.1(0.192) = 1.916 - 0.0192 = 1.8968$